

Exercise: LangGraph Multi-Agent Systems

Autonomous Marketing Campaign Squad

1. Scenario Overview

You are tasked with building an AI-driven marketing agency for a fast-paced startup. When given a simple product launch brief, a single AI struggles to do the research, write the copy, and verify the quality all at once without losing context or hallucinating.

Your goal is to design a **Multi-Agent Team** where specialized AI personas collaborate, review each other's work, and iteratively improve a social media campaign before presenting it to a human manager for final sign-off.

2. Business Objectives

By completing this exercise, you will successfully:

1. Orchestrate multiple specialized AI personas to collaborate on a single overarching goal.
2. Design a delegation and routing mechanism where a "manager" AI directs the flow of work.
3. Implement an evaluation and feedback loop to enforce autonomous quality control.
4. Establish a reliable safety guardrail by requiring human authorization before the final output is considered "published."

3. The Agent Team (Roles)

You need to create distinct instructions and personas for the following team members:

1. **The Campaign Manager (Supervisor):** * Reads the user's initial product brief.
 - Does not write the content itself. Instead, it decides which team member should act next based on the current progress of the campaign.
2. **The Trend Analyst (Worker):**
 - Simulates researching current market trends, relevant hashtags, and target audience pain points based on the product brief.
3. **The Copywriter (Worker):**
 - Takes the original brief and the Analyst's research to draft engaging social media posts.
4. **The QA Editor (Evaluator):**
 - Critiques the Copywriter's draft. It checks if the tone is appropriate, if the target audience is addressed, and if the post is free of clichés.
 - It has the authority to approve the draft or reject it with specific feedback.

4. The Required Workflow (The Flow)

Your multi-agent system should follow this operational logic:

1. **Initiation:** The user provides a product brief (e.g., *"We are launching a new eco-friendly smart water bottle. Target: gym-goers."*).
2. **Delegation Phase:** The Campaign Manager analyzes the request and routes the task to the Trend Analyst to gather context.
3. **Drafting Phase:** Once the research is complete, the Campaign Manager hands the context over to the Copywriter to produce the first draft.
4. **The Quality Loop:** * The Copywriter submits the draft to the QA Editor.
 - If the QA Editor finds the draft lacking, it sends it *back* to the Copywriter with a list of required changes (creating an autonomous revision cycle).
 - This cycle repeats until the QA Editor approves the content.
5. **The Human Guardrail:** Once the QA Editor approves the final draft, the system must completely pause. It cannot output the final campaign until a human reviewer reads the approved draft and gives the final "Go/No-Go" signal.
6. **Completion:** Upon human approval, the system outputs the finalized, ready-to-publish campaign.

5. Advanced Business Challenges

- **Budget/Time Constraints (Timeouts/Limits):** Ensure your QA Editor and Copywriter do not get stuck in an endless loop of revisions. Implement a rule where the system forces the workflow to end or escalate to a human if it exceeds a certain number of revision cycles.
- **Conflict Resolution:** If the human reviewer rejects the final draft at the guardrail stage, build a pathway for the human to provide manual feedback that sends the entire process back to the Copywriter for a fresh rewrite.